UE24CS341A SOFTWARE ENGINEERING · PES UNIVERSITY · JACKFRUIT MINI-PROJECT

Jira Product Backlog

Team 14 · Project 39: Dehallucination Add-On Module ("Dehallucinator")

Team member	Role
Sai Nikitha Perumalla (PES2UG24AM142)	Role A – Claim Extractor; Scrum Master
Shreya Raghuraj (PES2UG24AM154)	Role B – Evidence Retriever
Ralph D'souza (PES2UG24AM131)	Role C – Verifier & Scorer, testing, SonarQube
Satwik (PES2UG24AM148)	Role D – Corrector, API, UI, Jenkins, Docker

Repository: github.com/SoftwareEngineering-Runtime/dehallucinator · Jira project: DH
Exported from Jira project DH (Scrum board) on 07 Nov 2026
Date: 07 Nov 2026

1. Summary

The backlog has **11 epics** and **62 user stories and tasks** worth **162 story points**, planned over four sprints. Each story carries an owner (role A–D), story points (Fibonacci), the sprint it belongs to and its current status on the board (To Do → In Progress → In Review → Done). A story moves to Done when its pull request is reviewed, passes CI (ruff, pytest, SonarCloud) and is merged to `main`.

Points by sprint

Sprint	Stories	Points	Done points	Done %
Sprint 0 – Setup (18 Sep – 4 Oct 2026)	10	25	25	100%
Sprint 1 – Core (5 – 14 Oct 2026)	21	57	57	100%
Sprint 2 – Smart + connected (26 Oct – 4 Nov 2026)	19	56	27	48%
Sprint 3 – Validation + release (planned)	12	24	0	0%
Total	**62**	**162**	**109**	**67%**

Points by team member

Member	Stories	Points	Done points
Sai Nikitha Perumalla	18	44	23
Shreya Raghuraj	13	36	32
Ralph D'souza	15	38	25
Satwik	16	44	29

Epics

Epic	Name	Stories	Points
DH-141	E1 Project setup	5	14
DH-142	E2 Requirements (SRS)	5	11
DH-143	E3 Claim Extractor	5	18
DH-144	E4 Evidence Retriever	5	19
DH-145	E5 Verifier & Scorer	4	13
DH-146	E6 Corrector & Integration	5	16
DH-147	E7 CI/CD & Quality	4	13
DH-148	E8 Architecture & Design	9	17
DH-149	E9 Testing & Validation	9	21
DH-150	E10 Release	7	11
DH-209	E11 Scrum & project management	4	9

2. Sprint 0 – Setup (18 Sep – 4 Oct 2026)

Key	User story / task	Epic	Owner	Pts	Status
DH-151	As the team, we want a protected GitHub repo with the code skeleton so that everyone can start coding in parallel	E1	Sai Nikitha Perumalla	3	Done
DH-152	As the Scrum Master, I want Jira linked to GitHub so that commits show up on tickets	E1	Sai Nikitha Perumalla	1	Done
DH-153	As the team, we want a basic Jenkins pipeline so that every branch is linted and tested automatically	E1	Satwik	5	Done
DH-154	As the team, we want SonarQube Cloud analysis so that code quality is measured on every push	E1	Ralph D'souza	2	Done
DH-155	As the retriever, I want 15 trusted local KB documents so that evidence works offline	E1	Shreya Raghuraj	3	Done
DH-156	Write SRS section: UC-1 and FR-1.x (Claim Extractor)	E2	Sai Nikitha Perumalla	2	Done
DH-157	Write SRS section: Introduction, NFRs, UC-2 and FR-2.x (Evidence Retriever)	E2	Shreya Raghuraj	3	Done
DH-158	Write SRS section: UC-3, FR-3.x and validation approach	E2	Ralph D'souza	2	Done
DH-159	Write SRS section: UC-4, FR-4.x and external interfaces	E2	Satwik	2	Done
DH-160	Compile and export the final SRS	E2	Shreya Raghuraj	2	Done

3. Sprint 1 – Core (5 – 14 Oct 2026)

Key	User story / task	Epic	Owner	Pts	Status
DH-161	As the pipeline, I want input validated and split into sentences so that each sentence can be checked	E3	Sai Nikitha Perumalla	3	Done
DH-162	As the pipeline, I want non-factual sentences filtered out so that only checkable facts are verified	E3	Sai Nikitha Perumalla	3	Done
DH-166	As the retriever, I want a search query built for each claim so that Wikipedia returns relevant pages	E4	Shreya Raghuraj	3	Done
DH-167	As the retriever, I want to search Wikipedia and fetch page text so that claims can be checked against a trusted source	E4	Shreya Raghuraj	5	Done
DH-168	As the verifier, I want the most relevant evidence snippets ranked so that I verify against the best text	E4	Shreya Raghuraj	3	Done
DH-171	As the verifier, I want the NLI model loaded once so that verification is fast	E5	Ralph D'souza	3	Done
DH-172	As the verifier, I want each claim labelled SUPPORTED / CONTRADICTED / NOT_ENOUGH_INFO so that hallucinations are detected	E5	Ralph D'souza	5	Done
DH-173	As the user, I want an overall trust score so that I know how reliable the answer is	E5	Ralph D'souza	2	Done
DH-175	As the user, I want wrong claims corrected and unverified ones flagged so that I can trust what I read	E6	Satwik	3	Done
DH-176	As a developer, I want one check_answer() pipeline so that the modules work together	E6	Satwik	3	Done
DH-177	As a developer, I want a REST API so that any app can use the module	E6	Satwik	3	Done
DH-184	Architecture doc: overall diagram and data flow	E8	Sai Nikitha Perumalla	2	Done
DH-185	Architecture doc: Evidence Retriever component	E8	Shreya Raghuraj	1	Done
DH-186	Architecture doc: Verifier component	E8	Ralph D'souza	1	Done
DH-187	Architecture doc: API, UI and deployment diagram	E8	Satwik	2	Done
DH-193	Write test cases TC-1-01..10 for UC-1	E9	Sai Nikitha Perumalla	2	Done
DH-194	Write test cases TC-2-01..10 for UC-2	E9	Shreya Raghuraj	2	Done
DH-195	Write test cases TC-3-01..10 for UC-3 and compile the test plan	E9	Ralph D'souza	2	Done
DH-197	Integration tests for the full pipeline	E9	Ralph D'souza	3	Done
DH-210	Scrum: Sprint 1 planning, daily stand-ups and progress tracking	E11	Sai Nikitha Perumalla	3	Done
DH-211	Code review: review and merge Sprint 1 pull requests	E11	Sai Nikitha Perumalla	3	Done

4. Sprint 2 – Smart + connected (26 Oct – 4 Nov 2026)

Key	User story / task	Epic	Owner	Pts	Status
DH-163	As the pipeline, I want compound sentences split into separate claims so that each fact is verified on its own	E3	Sai Nikitha Perumalla	5	In Review
DH-164	As the retriever, I want leading pronouns replaced with the named entity so that claims are searchable on their own	E3	Sai Nikitha Perumalla	5	In Review
DH-165	As the pipeline, I want duplicate claims removed and claims numbered so that results are clean and traceable	E3	Sai Nikitha Perumalla	2	In Review
DH-169	As the retriever, I want to search the local KB so that the system works offline	E4	Shreya Raghuraj	3	Done
DH-170	As the retriever, I want caching and safe network handling so that it is fast and never crashes	E4	Shreya Raghuraj	5	Done
DH-174	As the verifier, I want wrong numbers and years caught so that numeric hallucinations are not missed	E5	Ralph D'souza	3	Done
DH-178	As a chatbot developer, I want a @dehallucinate decorator so that I can add the module to my bot in one line	E6	Satwik	2	Done
DH-179	As an end user, I want a demo chatbot UI so that I can see hallucinations highlighted and fixed	E6	Satwik	5	Done
DH-180	As the team, we want the full Jenkins pipeline so that every change is linted, scanned, tested and deployed	E7	Satwik	5	In Review
DH-181	As the team, we want a Docker image so that the app runs with one command	E7	Satwik	3	In Review
DH-182	As the team, we want bandit and pip-audit in CI so that security issues block the build	E7	Satwik	2	Done
DH-188	Design doc: class and sequence diagrams for the Claim Extractor	E8	Sai Nikitha Perumalla	2	In Review
DH-189	Design doc: class and sequence diagrams for the Evidence Retriever	E8	Shreya Raghuraj	2	Done
DH-190	Design doc: class and sequence diagrams for the Verifier	E8	Ralph D'souza	2	Done
DH-191	Design doc: diagrams for the Corrector, pipeline and API spec	E8	Satwik	3	In Review
DH-192	Compile and export the Architecture and Design documents	E8	Sai Nikitha Perumalla	2	In Progress
DH-196	Write test cases TC-4-01..10 for UC-4	E9	Satwik	2	Done
DH-212	Write Sprint 0 log (stand-ups, review, retrospective)	E11	Sai Nikitha Perumalla	1	Done
DH-213	Sprint 1 review and retrospective	E11	Sai Nikitha Perumalla	2	In Progress

5. Sprint 3 – Validation + release (planned)

Key	User story / task	Epic	Owner	Pts	Status
DH-183	As the team, we want the SonarQube quality gate passing so that maintainability is proven	E7	Ralph D'souza	3	To Do
DH-198	System tests on the 10 sample answers	E9	Ralph D'souza	3	To Do
DH-199	Security tests	E9	Ralph D'souza	3	To Do
DH-200	Validation report	E9	Ralph D'souza	3	To Do
DH-201	Coverage report export	E9	Satwik	1	To Do
DH-202	Maintenance plan	E10	Satwik	2	To Do
DH-203	README: project overview, setup and Claim Extractor section	E10	Sai Nikitha Perumalla	1	To Do
DH-204	README: Evidence Retriever section	E10	Shreya Raghuraj	1	To Do
DH-205	README: Verifier section	E10	Ralph D'souza	1	To Do
DH-206	README: API, UI and deployment section	E10	Satwik	1	To Do
DH-207	Final report and presentation slides	E10	Shreya Raghuraj	3	To Do
DH-208	Demo rehearsals and backup recording	E10	Sai Nikitha Perumalla	2	To Do